

Inho Choi

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EDUCATION

- **National University of Singapore** Singapore
Ph.D. Candidate in School of Computing (Advisor: Jialin Li); August 2019 – December 2025 (Expected)
- **Yonsei University** Seoul, Republic of Korea
Bachelor's Degree in Computer Science; March 2012 – August 2019
- **Uppsala University** Uppsala, Sweden
Exchange Student in Information Technology; August 2017 – January 2018

WORK IN PROGRESS *(projects that I'm currently leading)*

- **ML-native Dataplane Operating Systems** [[Current Main Project](#)]

We design an ML-native dataplane OS architecture for automatic parameter tuning. Performance tuning remains a persistent challenge in modern datacenters, especially at microsecond scales. We are exploring how machine learning can be natively integrated into the OS dataplane, with case studies showing substantial performance gains through dynamic parameter optimization. Through this work, I actively perform comprehensive full-stack performance analysis and optimization, spanning from hardware-level signals (PCIe, memory access patterns, cache efficiency, DDIO) and inter-device data exchange to device drivers, OS components, and application layers.

- **Dynamic Layer-4 Load Balancing with Microsecond-Scale TCP Migration** [[Under Review](#)]

Layer-4 load balancers are a popular solution to high tail latencies but perform poorly under unpredictable workloads and traffic bursts because they statically assign connections to servers. We propose the first dynamic L4 load balancer with μ s-scale stateful connection migration. Our system leverages two trends – programmable switches and kernel-bypass – to efficiently implement TCP migration without packet loss, while maintaining transparency to clients.

PUBLICATIONS

- **[APSYS '25] ML-native Dataplane Operating Systems:**
Inho Choi, Anand Bonde, Jing Liu, Joshua Fried, Irene Zhang, Jialin Li.
16th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2025)
- **[ArXiv] A Primer on RecoNIC: RDMA-enabled Compute Offloading on SmartNIC:**
Guanwen Zhong, Aditya Kolekar, Burin Amornpaisannon, Inho Choi, Haris Javaid, Mario Baldi.
ArXiv, Dec, 2023
- **[APSYS '23] Capybara: μ Second-scale live TCP migration:**
Inho Choi, Nimish Wadekar, Raj Joshi, Joshua Fried, Dan R. K. Ports, Irene Zhang, and Jialin Li.
14th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2023)
- **[SIGCOMM '23] Network Load Balancing with In-network Reordering Support for RDMA:**
Cha Hwan Song, Xin Zhe Khooi, Raj Joshi, Inho Choi, Jialin Li, and Mun Choon Chan.
Proceedings of the 2023 ACM SIGCOMM Conference
- **[NSDI '23] Hydra: Serialization-Free Network Ordering for Strongly Consistent Distributed Applications:**
Inho Choi, Ellis Michael, Yunfan Li, Dan Ports, and Jialin Li.
Proceedings of the 20th USENIX Conference on Network Systems Design and Implementation

- **[S&P (Oakland) '20] A Stealthier Partitioning Attack against Bitcoin Peer-to-Peer Network:**
Muoi Tran, **Inho Choi**, Gi Jun Moon, Viet-Anh Vu, and Min Suk Kang.
In Proceedings of IEEE Symposium on Security and Privacy, May 2020.
- **[UbiComp Workshop '17] Multimodal Data Collection Framework for Mental Stress Monitoring:**
Saewon Kye, Junhyung Moon, Juneil Lee, **Inho Choi**, Dongmi Cheon, Kyoungwoo Lee.
In Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers. (Workshop Paper)

EXPERIENCES

- **Microsoft Research - PhD Research Intern** Redmond, WA, USA
Systems Research Group (Mentor: Irene Zhang) *June 2025 — September 2025*
Conducted research on ML-native dataplane OS architecture for automatic parameter tuning to address performance optimization challenges in modern datacenters at microsecond scales. Designed system architecture integrating machine learning natively into the OS dataplane for dynamic parameter optimization. Developed and evaluated the approach to demonstrate benefits of automated tuning across diverse workloads. Published initial findings at APSys '25, with full-system development currently in progress.
- **Microsoft Research - PhD Research Intern** Redmond, WA, USA
Systems Research Group (Mentor: Dan Ports) *June 2024 — August 2024*
Conducted research on dynamic Layer-4 load balancing to address tail latency issues in unpredictable workloads and traffic bursts caused by static TCP connection assignment between client and server. Designed and implemented microsecond-scale TCP connection migration system leveraging programmable switches and kernel-bypass technologies, achieving packet-loss-free migration while maintaining client transparency. Published initial findings at APSys '23, with full paper currently under submission.
- **AMD - PhD Research Intern** Singapore
Xilinx Lab - FPGA / System Design (Mentor: Guanwen Zhong) *May 2023 - August 2023*
Contributed to research on hardware accelerator architectures for datacenter networks, specializing in RDMA protocol optimization and computation offloading on FPGA-based SmartNIC platforms. Participated in developing offloading techniques to enhance network stack performance while minimizing host CPU overhead. Participated in implementing and validating solutions through FPGA prototype and performance evaluation, with findings published on ArXiv.
- **National University of Singapore - Research Intern** Singapore
Systems & Network Security Lab (Advisor: Min Suk Kang) *September 2018 - February 2019*
Contributed to research on network-level security vulnerabilities in Bitcoin's peer-to-peer network, analyzing how adversarial autonomous systems can execute stealth attacks without routing manipulations. Participated in developing and evaluating the EREBUS attack, demonstrating partition vulnerabilities in Bitcoin nodes and defense mechanisms through protocol modifications. Co-authored paper published at IEEE S&P '20, advancing cryptocurrency network security.
- **Metlife - Summer Intern** Seoul, Korea
IT Planning Team *July 2018 - August 2018*
Analyzed IT infrastructure and database server architecture at MetLife Korea, investigating OAuth 2.0 protocol adoption for security enhancement. Presented findings through internal seminar.
- **Yonsei University - Research Intern** Seoul, Korea
Dependable Computing Lab (Advisor: Kyoungwoo Lee) *February 2017 - May 2018*
Contributed to development of a multimodal stress monitoring framework for analyzing people's physiological and behavioral reactions to stressors. Participated in designing and conducting experiments and implementing real-time signal processing framework. Co-authored paper published at UbiComp Workshop '17, presenting the framework and experimental findings.

AWARDS

- **Research Incentive Award** October 2023
National University of Singapore
- **NUS Research Scholarship** August 2019 – July 2023
National University of Singapore
- **Research Achievement Award** January 2023
National University of Singapore
- **Honors - 1st Semester, 2018** August 2018
Yonsei University

TEACHING EXPERIENCES

- **Teaching Assistant for Distributed Systems (CS5223)** Semester 1, 2022
National University of Singapore
- **Teaching Assistant for Introduction to Operating Systems (CS2106)** Semester 1, 2021
National University of Singapore
- **Teaching Assistant for Distributed Systems (CS5223)** Semester 2, 2020
National University of Singapore
- **Teaching Assistant for Programming Methodology (CS1010E)** Semester 1, 2020
National University of Singapore
- **Teaching Assistant for Cryptography Theory and Practice (CS4236)** Semester 1, 2019
National University of Singapore

MENTORING EXPERIENCES

- **Yiyang Liu — NUS, Singapore** 2024
Master's Thesis: Enhancing Distributed Systems with Hydra: A Software Solution for Scalable Network Ordering

SERVICES

- **Reviewer** ToN '25
- **Shadow PC Reviewer** EuroSys '26, EuroSys '25
- **Student Volunteer** SOSP '21

INVITED TALKS

- **Co-Designing Systems for Microsecond-Scale Consistent Tail Latency in Datacenters**
Yonsei University (2025), Seoul National University (2025)